Momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate and compare signed momentum.

SUCCESS CRITERIA - I can...

1. Multiply mass by velocity with correct units.
2. Include direction or a sign.
3. Distinguish zero total momentum from no motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “velocity” to compare your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

velocity: rate of position change, with direction

mass: a measure of how strongly an object resists acceleration

01 Bring back an earlier idea

An asteroid’s period is 5 years.
Explain why that number alone cannot establish an Earth encounter.
Name two additional observations needed.

Word help: encounter = objects reaching nearby positions at the same time

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $p_{total} = p_1 + p_2$.
Use signed velocities; momentum units are kg·m/s.

Worked reasoning: A 3 kg cart moving left at 2 m/s has momentum -6 kg·m/s when right is positive. The minus sign gives direction, not a negative mass.

Your turn: Draw separate velocity and momentum arrows, labeling their different units.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

03 Look and explain

The two carts have the same speed in opposite directions. Which momentum values are positive if right is positive?

Sentence starter: I notice … . This supports … because … .



2 kg; 3 m/s right

2 kg; 3 m/s left

Marzano check - circle: ① ② ③

04 Build the solution

Calculate each momentum in the picture and their total. Does zero total mean the carts are not moving?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

One cart doubles mass at the same velocity. Another doubles speed at the same mass.
Compare the momentum multipliers.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg test mass feels 24 N toward a source.
Find g and the force on a 8 kg mass at the same point.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A 4 kg cart moves left at 2 m/s.
State its momentum with direction.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Build two moving-cart examples whose total momentum is zero but whose masses differ.
Draw both momentum arrows.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3